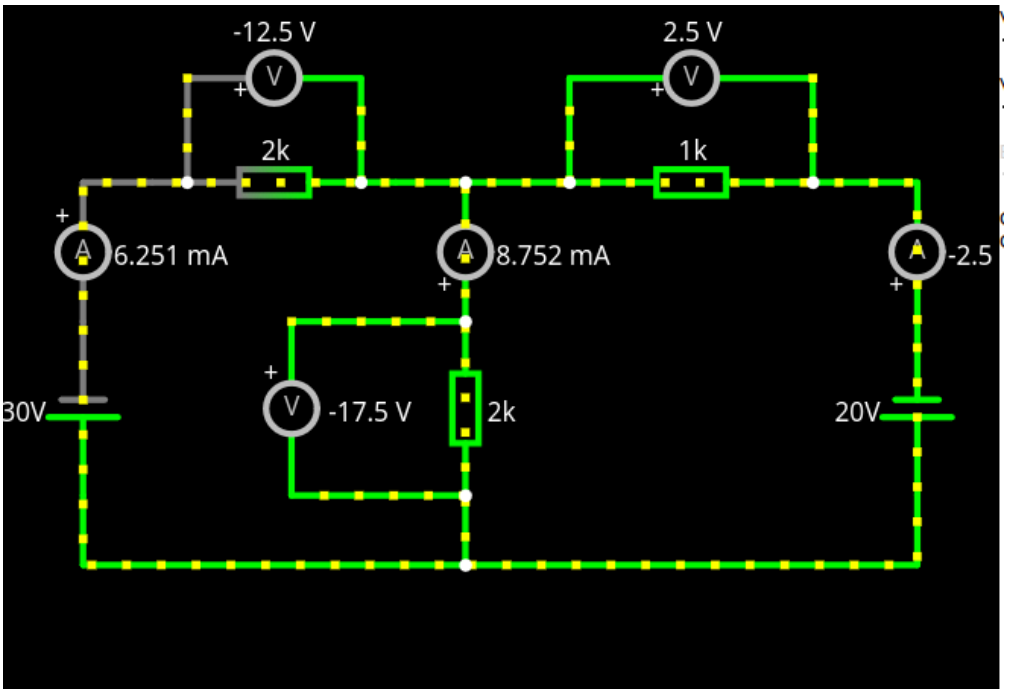
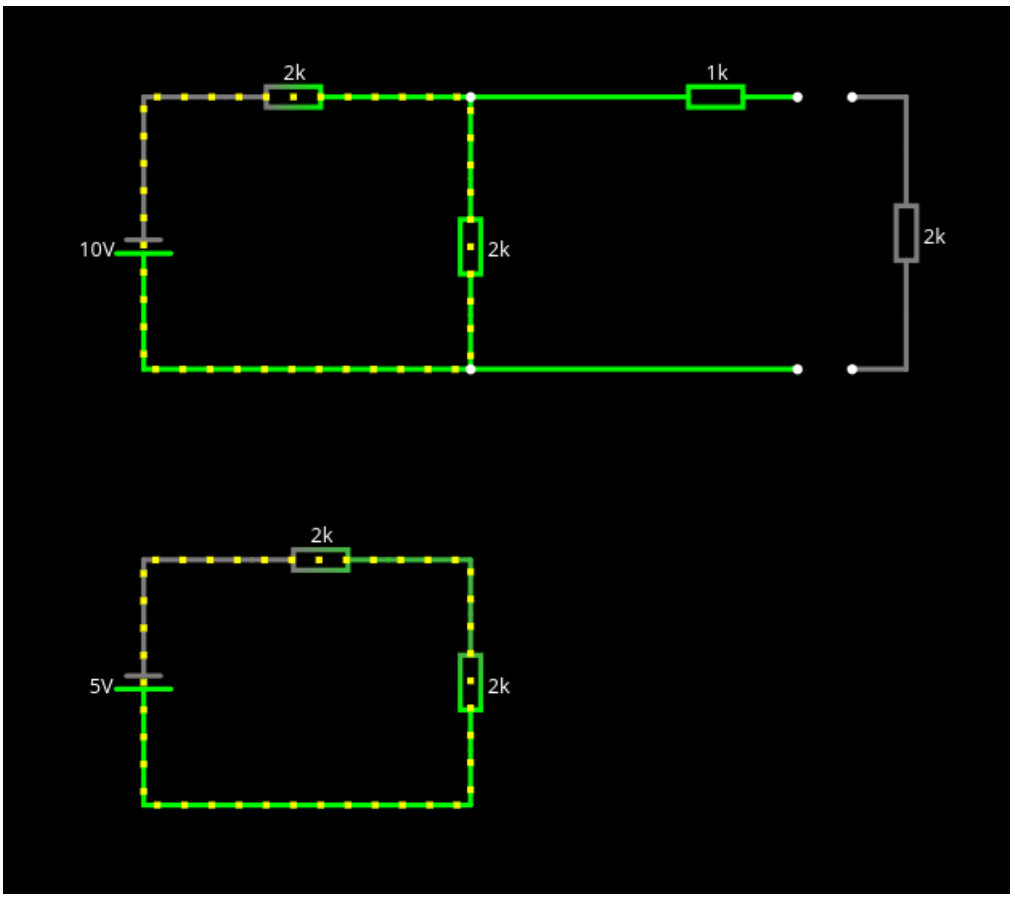
Exemplo de Sobreposição

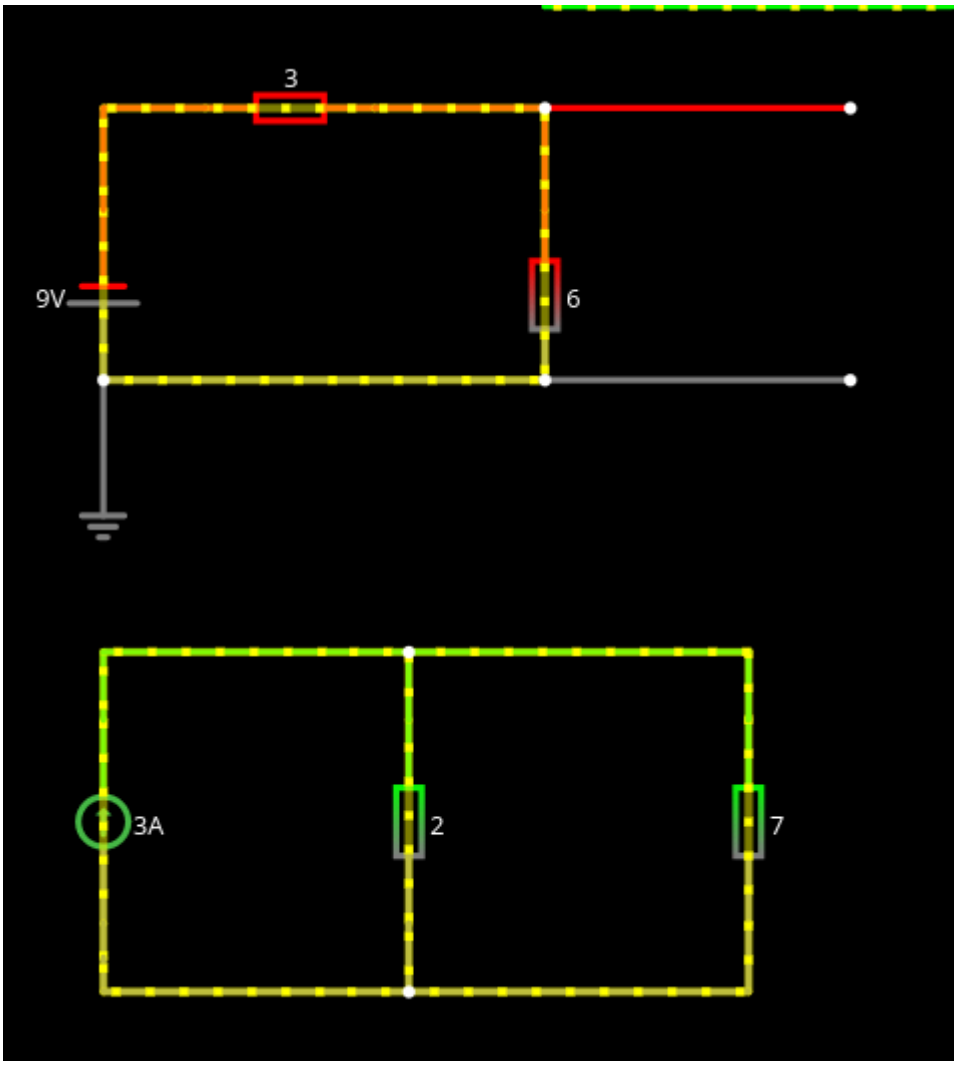


Exemplo de Thévenin



CamScanner

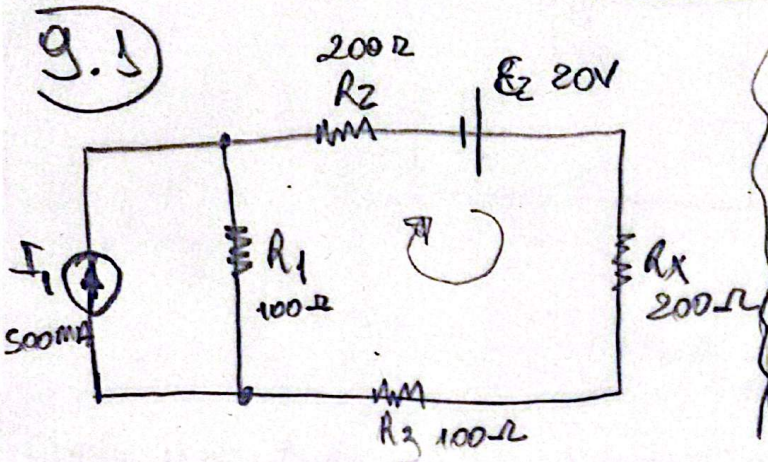
Exemplo de Norton



CamScanner

Atividades
Sobreposição, Thévenin e Norton
(Analise circuitos cc)

9.1)



I_x e $\mathcal{E}_x$?

Fonte de corrente I_1:

$R_t = R_2 + R_x + R_3 = 500\Omega$

Corrente em I_x:

$$I_x' = I_1 \cdot \left(\frac{R_1}{R_1 + R_t}\right) = 0{,}5 \cdot \frac{100}{100+500}$$

$$= 0{,}5 \cdot \frac{100}{600} = \frac{1}{12} A \approx 83{,}3mA \text{ (para baixo)}$$

tensão em R_x:

$$V_x' = I_x' \cdot R_x = \frac{1}{12} \cdot 200 = \frac{50}{3} V \approx 16{,}67V$$

Fonte de tensão:

Resist. total

$$R_t = R_1 + R_2 + R_x + R_3 = 600\Omega$$

Corrente em I_x'':

$$I_x'' = -\frac{\mathcal{E}_2}{R_t} = \frac{-200}{600} = -\frac{1}{50} \simeq -33{,}3mA$$

tensão em V_x'':

$$V_x'' = I_x'' \cdot R_x = -\frac{1}{30} \cdot 200 = -\frac{20}{3} V \simeq -6{,}67V$$

Resultados finais (Sobreposição)

Corrente total:

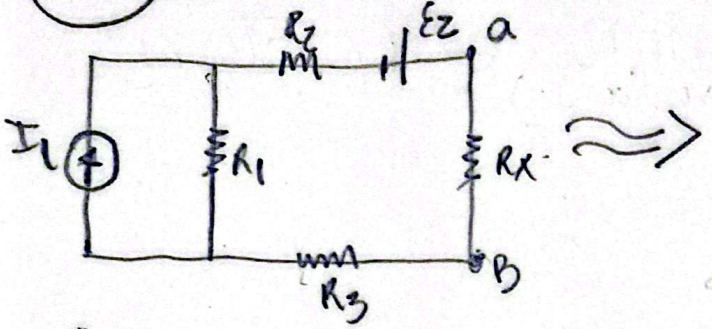
$$I_x = I_x' + I_x'' = \frac{1}{12} - \frac{1}{30} = \frac{3}{60} = 0{,}05A \text{ ou } 50mA$$

tensão total:

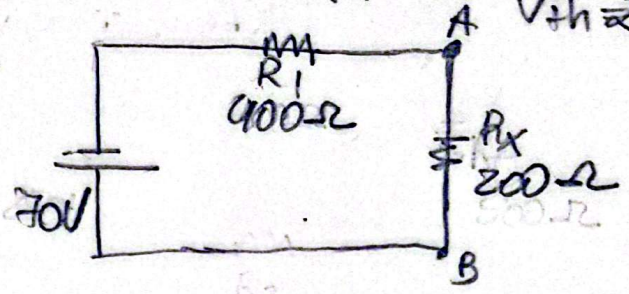
$$V_x = V_x' + V_x'' = \frac{50}{3} - \frac{20}{3} = \frac{30}{3} = 10V$$

CamScanner

9.2



$\Rightarrow$



$I_x = \frac{V_{th}}{R_{th}} \Rightarrow 0{,}5A = \frac{V_{th}}{500}$

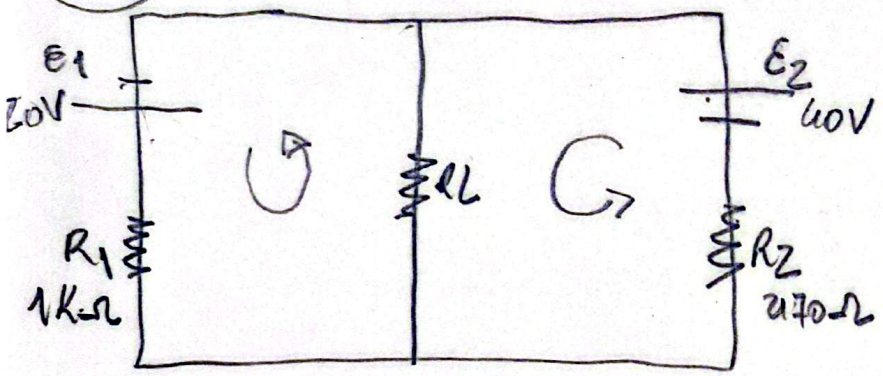
$V_{th} = 250V$

$\mathcal{E}_1 = I_1 \cdot R_1 = 0{,}5A \cdot 100 = 50V$

$Req\,th = R_1 + R_2 + R_3 = 400\Omega$

$V_{th} = \mathcal{E}_1 + \mathcal{E}_2 = 50V + 20V = 70V$

9.3



Qual o valor de I_L e V_L em:

$R_L = 100\Omega = 500\Omega = 1500\Omega$

Resistencia equivalente de R1 e R2

$R_{th} = \frac{1000 \cdot 470}{1000 + 470} \simeq 319{,}73\Omega$

$V_{th} = \frac{-\frac{\mathcal{E}_1}{R_1} + \frac{\mathcal{E}_2}{R_2}}{\frac{1}{R_1} + \frac{1}{R_2}} = 20{,}82V$

Para cada carga R_L, veremos essas formulas:

$I_L = \frac{V_{th}}{R_{th} + R_L}$ e $V_L = I_L \cdot R_L$

a) $R_{L1} = 100\Omega$

$I_{L1} = \frac{20{,}82}{319 + 100} = \frac{20{,}82}{419{,}73} \simeq 49{,}60mA$ $\Rightarrow V_{L1} = 99{,}60mA \cdot 100 = 4{,}96V$

b) $R_{L2} = 500$

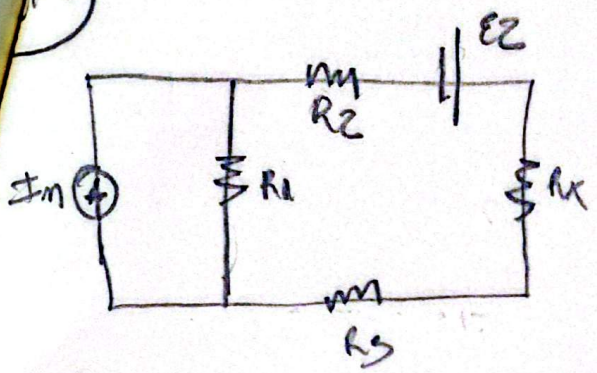
$I_{L2} = \frac{20{,}82}{319{,}73 + 500} = \frac{20{,}82}{819{,}73} \simeq 25{,}4mA$ $\Rightarrow V_{L2} = 25{,}4mA \cdot 500\Omega \simeq 12{,}7V$

c) $R_{L3} = 1500\Omega$

$I_{L2} = \frac{20{,}82}{319{,}73 + 1500} = \frac{20{,}82}{1819{,}73} \simeq 11{,}44mA$ $\Rightarrow V_{L3} = 11{,}44 \cdot 1500 \simeq 17{,}16V$

CamScanner

9.4



$R_N = R_1 + R_2 + R_3 = 400\Omega$

$\mathcal{E}'_1 = I_1 \cdot R_1 = 0{,}5A \cdot 100\Omega = 50V$

$\mathcal{E}_{total} = \mathcal{E}'_1 + \mathcal{E}_2 = 50 + 20 = 70V$

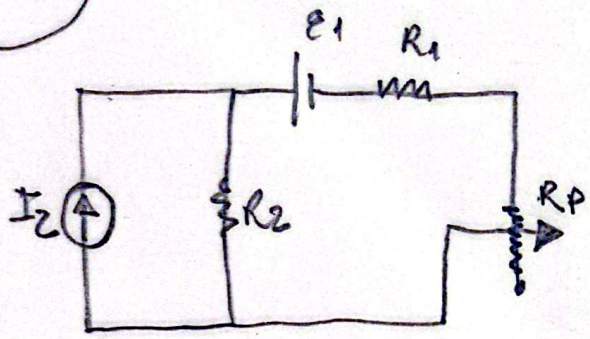
$I_N = \frac{\mathcal{E}_{total}}{R_N} = \frac{70}{400} = 175mA$

Logo...

$I_x = I_N \cdot \frac{R_N}{R_N + R_x} = 175mA \frac{400}{400+200} \cong 116{,}67mA$

$U_x = I_x \cdot R_x = \frac{7}{60} A \cdot 200 \cong 23{,}3V$

9.5



Resistencia de Norton (R_N)

$R_N = R_1 + R_2$

$R_N = 820 + 1200 = 2020\Omega$

Corrente de Norton (I_N)

$\mathcal{E}'_2 = I_2 \cdot R_2 = 0{,}01A \cdot 1200 = 48V$

$\mathcal{E}_{eq} = \mathcal{E}'_2 - \mathcal{E}_1 = 48V - 24V = 24V$

$I_N = \frac{\mathcal{E}_{eq}}{R_N} = \frac{24}{2020} \cong 11{,}881mA$

a) Para $R_P = 0\Omega$

$I_{RP} = \frac{24}{2020+0} \cong 11{,}88mA$

b) Para $R_P = 600\Omega$

$I_{RP} = \frac{24}{2020+600} = \frac{24}{2620} \cong 9{,}16mA$

c) Para $R_P = 1200\Omega$

$I_{RP} = \frac{24}{2020+1200} = \frac{24}{3220} \cong 7{,}45mA$

D) Para $R_P = 1800\Omega$

$I_{RP} = \frac{24}{2020+1800} = \frac{24}{3820} \cong 6{,}28mA$

E) Para $R_P = 1800\Omega$

$I_{RP} = \frac{24}{2020+1800} = \frac{24}{4220} \cong 5{,}69mA$

CamScanner